

《数据库系统原理》实验报告（）																																																																	
题目：DDL 语言实验																																																																	
学号	2351136	姓名	李盛鹏	日期	2025.4.6																																																												
实验环境： 硬件环境： windows\wsl, Docker Desktop 版本 28.0.1 软件环境： OceanBase 社区版镜像:oceanbase/oceanbase-ce:lastest																																																																	
实验步骤及结果截图： 1. 创立表并查询表结构 1.1 建立 Customers 并查询 <pre>CREATE TABLE Customers (CustomerID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Name VARCHAR(50) NOT NULL, Email VARCHAR(100) NOT NULL UNIQUE CHECK (Email LIKE '%@%.' AND Email NOT LIKE '%..%' AND Email NOT LIKE '%@.%' AND Email NOT LIKE '%.%@'), Phone VARCHAR(20) CHECK (Phone REGEXP '^([0-9]{10,20})\$'), -- 假设电话号码由10到 20位数字组成 RegistrationDate DATETIME not null DEFAULT CURRENT_TIMESTAMP);</pre> <pre>obclient(root@sys)[mamba]> desc Customers;</pre> <table><tr><th>Field</th><th>Type</th><th>Null</th><th>Key</th><th>Default</th><th>Extra</th></tr><tr><td>CustomerID</td><td>int(11)</td><td>NO</td><td>PRI</td><td>NULL</td><td>auto_increment</td></tr><tr><td>Name</td><td>char(15)</td><td>NO</td><td></td><td>NULL</td><td></td></tr><tr><td>Email</td><td>varchar(100)</td><td>NO</td><td>UNI</td><td>NULL</td><td></td></tr><tr><td>Phone</td><td>varchar(20)</td><td>YES</td><td></td><td>NULL</td><td></td></tr><tr><td>RegistrationDate</td><td>datetime</td><td>NO</td><td></td><td>CURRENT_TIMESTAMP</td><td></td></tr><tr><td>LastLoginDate</td><td>datetime</td><td>YES</td><td></td><td>NULL</td><td></td></tr></table> 6 rows in set (0.006 sec) 1.2 创立 Categories 表并查询 <pre>CREATE TABLE Categories (CategoryID INT NOT NULL AUTO_INCREMENT PRIMARY KEY, CategoryName VARCHAR(50) NOT NULL UNIQUE);</pre> <pre>obclient(root@sys)[mamba]> desc Categories;</pre> <table><tr><th>Field</th><th>Type</th><th>Null</th><th>Key</th><th>Default</th><th>Extra</th></tr><tr><td>CategoryID</td><td>int(11)</td><td>NO</td><td>PRI</td><td>NULL</td><td>auto_increment</td></tr><tr><td>CategoryName</td><td>varchar(50)</td><td>NO</td><td>UNI</td><td>NULL</td><td></td></tr></table> 1.3 创立 Products 表并查询						Field	Type	Null	Key	Default	Extra	CustomerID	int(11)	NO	PRI	NULL	auto_increment	Name	char(15)	NO		NULL		Email	varchar(100)	NO	UNI	NULL		Phone	varchar(20)	YES		NULL		RegistrationDate	datetime	NO		CURRENT_TIMESTAMP		LastLoginDate	datetime	YES		NULL		Field	Type	Null	Key	Default	Extra	CategoryID	int(11)	NO	PRI	NULL	auto_increment	CategoryName	varchar(50)	NO	UNI	NULL	
Field	Type	Null	Key	Default	Extra																																																												
CustomerID	int(11)	NO	PRI	NULL	auto_increment																																																												
Name	char(15)	NO		NULL																																																													
Email	varchar(100)	NO	UNI	NULL																																																													
Phone	varchar(20)	YES		NULL																																																													
RegistrationDate	datetime	NO		CURRENT_TIMESTAMP																																																													
LastLoginDate	datetime	YES		NULL																																																													
Field	Type	Null	Key	Default	Extra																																																												
CategoryID	int(11)	NO	PRI	NULL	auto_increment																																																												
CategoryName	varchar(50)	NO	UNI	NULL																																																													

```
create table Products(
  ProductID INT not null AUTO_INCREMENT,
  ProductName VARCHAR(100) not null,
  Price DECIMAL(10,2) not null check (Price > 0),
  StockQuantity INT not null check(StockQuantity >= 0),
  CategoryID INT not null,
  PRIMARY KEY(ProductID),
  FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID));
```

```
obclient(root@sys)[mamba]> desc Products;
```

Field	Type	Null	Key	Default	Extra
ProductID	int(11)	NO	PRI	NULL	auto_increment
ProductName	varchar(100)	NO		NULL	
Price	decimal(10,2)	NO		NULL	
StockQuantity	int(11)	NO		NULL	
CategoryID	int(11)	NO		NULL	
IsFeatured	tinyint(1)	YES		0	

1.4 创立 Orders 表并查询

```
create table Orders(
  OrderID INT AUTO_INCREMENT not null,
  CustomerID INT not null,
  OrderDate DATETIME not null DEFAULT CURRENT_TIMESTAMP,
  TotalAmount DECIMAL(10,2) not null check(TotalAmount > 0),
  PRIMARY KEY(OrderID),
  FOREIGN KEY(CustomerID) REFERENCES Categories(CategoryID)
);
```

```
obclient(root@sys)[mamba]> desc Orders;
```

Field	Type	Null	Key	Default	Extra
OrderID	int(11)	NO	PRI	NULL	auto_increment
CustomerID	int(11)	NO		NULL	
OrderDate	datetime	NO		CURRENT_TIMESTAMP	
TotalAmount	decimal(10,2)	NO		NULL	

4 rows in set (0.006 sec)

1.5 创立 OrderItems 表并查询

```
CREATE TABLE OrderItems (
  OrderItemID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
  OrderID INT NOT NULL,
  ProductID INT NOT NULL,
  Quantity INT NOT NULL CHECK (Quantity > 0),
  Subtotal DECIMAL(10,2) NOT NULL CHECK (Subtotal > 0),
  FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
  FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);
```

```
CREATE TRIGGER before_insert_order_items
BEFORE INSERT ON OrderItems
FOR EACH ROW
BEGIN
    DECLARE product_price DECIMAL(10,2);
    SELECT Price INTO product_price FROM Products WHERE ProductID = NEW.ProductID;
    SET NEW.Subtotal = product_price * NEW.Quantity;
END //

CREATE TRIGGER before_update_order_items
BEFORE UPDATE ON OrderItems
FOR EACH ROW
BEGIN
    DECLARE product_price DECIMAL(10,2);
    SELECT Price INTO product_price FROM Products WHERE ProductID = NEW.ProductID;
    SET NEW.Subtotal = product_price * NEW.Quantity;
END //
```

触发器

```
obclient(root@sys)[mamba]> desc OrderItems;
```

Field	Type	Null	Key	Default	Extra
OrderItemID	int(11)	NO	PRI	NULL	auto_increment
OrderID	int(11)	NO		NULL	
ProductID	int(11)	NO		NULL	
Quantity	int(11)	NO		NULL	
Subtotal	decimal(10,2)	NO		NULL	

5 rows in set (0.006 sec)

2. 查询建立的约束

2.1 Customers 表的约束查询

```
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name='Customers';
```

table_name	constraint_name	constraint_type
Customers	PRIMARY	PRIMARY KEY
Customers	Email	UNIQUE
Customers	Customers_OBCHECK_1743862686701471	CHECK
Customers	Customers_OBCHECK_1743862686701697	CHECK

2.2 Products 表的约束查询

```
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name='Products';
```

table_name	constraint_name	constraint_type
Products	PRIMARY	PRIMARY KEY
Products	Products_OBCHECK_1743862703465322	CHECK
Products	Products_OBCHECK_1743862703465542	CHECK
Products	Products_OBFK_1743862703465619	FOREIGN KEY

2.3 Categories 表的约束查询

```
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type
WHERE constraint_name='Categories';
```

table_name	constraint_name	constraint_type
Categories	PRIMARY	PRIMARY KEY
Categories	CategoryName	UNIQUE

2.4 Orders 表的约束查询

```
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type
WHERE constraint_name='Orders';
```

table_name	constraint_name	constraint_type
Orders	PRIMARY	PRIMARY KEY
Orders	Orders_OBCHECK_1743862710517010	CHECK
Orders	Orders_OBFK_1743862710517118	FOREIGN KEY

3 rows in set (0.005 sec)

2.5 OrderItems 表的约束查询

```
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type
WHERE constraint_name='OrderItems';
```

table_name	constraint_name	constraint_type
OrderItems	PRIMARY	PRIMARY KEY
OrderItems	OrderItems_OBCHECK_1743863976532941	CHECK
OrderItems	OrderItems_OBCHECK_1743863976533054	CHECK
OrderItems	OrderItems_OBFK_1743863976533419	FOREIGN KEY
OrderItems	OrderItems_OBFK_1743863976533297	FOREIGN KEY

5 rows in set (0.013 sec)

3.在 Customers 表中添加一个新字段 LastLoginDate，并自动更新为当前时间

```
obclient(root@sys)[mamba]> alter table Customers
-> ADD LastLoginDate DATETIME;
Query OK, 0 rows affected (0.091 sec)
```

```
obclient(root@sys)[mamba]> desc Customers
-> ;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int(11)	NO	PRI	NULL	auto_increment
Name	varchar(50)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Phone	varchar(20)	YES		NULL	
RegistrationDate	datetime	NO		CURRENT_TIMESTAMP	
LastLoginDate	datetime	YES		NULL	

```
obclient(root@sys)[mamba]> create trigger update_LastLoginDate
-> before update on Customers
-> For each row
-> set NEW.LastLoginDate = NOW();
Query OK, 0 rows affected (0.044 sec)
```

4. 将 Customers 中的 name 由 varchar 改为 char(15)


```
obclient(root@sys)[mamba]> alter table Customers
-> MODIFY Name CHAR(15) not null;
Query OK, 0 rows affected (0.567 sec)
```

```
obclient(root@sys)[mamba]> desc Customers;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int(11)	NO	PRI	NULL	auto_increment
Name	char(15)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Phone	varchar(20)	YES		NULL	
RegistrationDate	datetime	NO		CURRENT_TIMESTAMP	
LastLoginDate	datetime	YES		NULL	

5. 添加一个 IsFeatured 字段，并且创建一个索引，用于快速查询特色商品

```
obclient(root@sys)[mamba]> CREATE INDEX idx_products_featured ON Products((CASE WHEN IsFeatured = TRUE THEN 1 ELSE NULL END));
Query OK, 0 rows affected (0.266 sec)
```

6. 删除 Categories 表中所有没有关联商品的分类记录，创建一个存储过程实现此功能

```
obclient(root@sys)[mamba]> CREATE PROCEDURE DeleteUnrelatedCategories()
-> BEGIN
->     DELETE FROM Categories
->     WHERE CategoryID NOT IN (SELECT CategoryID FROM Products);
-> END //
Query OK, 0 rows affected (0.033 sec)
```

出现的问题：

1. 索引无法建立

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near 'WHERE IsFeatured = TRUE' at line 1
[172.17.0.2:2882] [2025-04-05 16:12:20.789503] [YB42AC110002-000632089E8FB4C3-0-0]
obclient(root@sys)[mamba]> CREATE INDEX idx_products_featured ON Products((CASE WHEN IsFeatured = TRUE THEN 1 ELSE NULL END));
```

2. mysql 关键词错误

```
obclient(root@sys)[mamba]> alter table Products
-> ADD IsFeatured BOOLEAN DEAFULT FALSE;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near 'DEAFULT FALSE' at line 2
```

3. Oceanbase 函数不支持

```
-> RegistrationDate DATETIME NOT NULL DEFAULT CURRENT_DATE);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near 'CURRENT_DATE' at line 6
[172.17.0.2:2882] [2025-04-05 14:15:39.002747] [YB42AC110002-000632089D0FB4AA-0-0]
```

4. 未添加分号

```
ERROR 1054 (42S22): Unknown column 'Products' in 'where clause'
[172.17.0.2:2882] [2025-04-05 15:52:52.461386] [YB42AC110002-000632089E8FB4B6-0-0]
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type FROM information_schema.TABLE_CONSTRAINTS WHERE table_name= 'Products';
```

解决方案:

1. 使用函数索引

```
obclient(root@sys)[mamba]> CREATE INDEX idx_products_featured ON Products((CASE WHEN IsFeatured = TRUE THEN 1 ELSE NULL END));
Query OK, 0 rows affected (0.266 sec)
```

2. 检查 default, 拼读重写

3. 更换类型为 Datetime, 缺省为 timestamp

```
obclient(root@sys)[mamba]> CREATE TABLE Customers (
->   CustomerID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
->   Name VARCHAR(50) NOT NULL,
->   Email VARCHAR(100) NOT NULL UNIQUE CHECK (Email LIKE '%@%.' AND Email NOT LIKE '%..%' AND Email NOT LIKE '%@.%' AND E
mail NOT LIKE '%.%@'),
->   Phone VARCHAR(20) CHECK (Phone REGEXP '^[0-9]{10,20}$'), -- 假设电话号码由10到20位数字组成
->   RegistrationDate DATETIME not null DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.088 sec)
```

4. 补充分号

```
obclient(root@sys)[mamba]> SELECT table_name, constraint_name, constraint_type FROM information_schema.TABLE_CONSTRAINTS WHERE ta
ble_name= 'Products';
```

table_name	constraint_name	constraint_type
Products	PRIMARY	PRIMARY KEY
Products	Products_OBCHECK_1743862703465322	CHECK
Products	Products_OBCHECK_1743862703465542	CHECK
Products	Products_OBFK_1743862703465619	FOREIGN KEY